



Unmanned aerial vehicles applications in future smart cities

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ABSTRACT

Foreseeing changes in how smart cities manage their resources and provide services to the residents; research, development and production in various relevant technology fields is accelerating. Taking advantage of recent advances and innovations in Information and Communication Technologies (ICT), robotics and software; smart cities can optimize resources utilization and enhance operations in health, transportation, energy, and water services, as well as elevating the level of comfort of residents. Effectively and efficiently utilizing ICT and robotics in smart cities will result in reducing costs and resources consumption in addition to engaging more effectively and actively with the citizens. One of these technologies is the unmanned aerial vehicle (UAV), which can provide many applications for smart cities and create a positive impact on the society. For example, UAVs can be used for environmental monitoring, traffic management, pollution monitoring, civil security control, and merchandise delivery. UAV applications among several others can provide cost-effective services to help achieve the objectives of smart cities. However, the integration of UAVs in smart cities is very challenging due to several issues and concerns such as safety, privacy and ethical/legal use. This paper reviews the potential applications integrating UAVs in smart cities, their implications, and the technical and non-technical issues facing such integration. It also discusses regulations and enabling technologies currently available and being developed that can be utilized to support such integration.

1. Introduction

The world population is rapidly increasing and expected to double by 2050. There is also the continuous where many people are moving from rural areas to the cities. This trend is anticipated to grow more rapidly in the coming years especially in developing countries. This rapid increase in population in cities will create many challenges for cities' administrators striving to maintain or enhance the cities' services and the residents' quality of life. Consequently, there is an increased interest in utilizing advanced Information and Communication Technologies (ICT), robotics and smart solutions in smart cities developments (Shreih, 2013). These technologies will help create smart automated services to improve the performance of infrastructures and comfort level of residents. The main goal of a smart city design is to provide efficient infrastructures and services at reduced costs. The European Platform for Intelligent Cities and the European Network of Living Labs defined smart cities as "The use of discrete new technology applications such as RFID and Internet of Things through more holistic conception of intelligent, integrated working that is closely linked to

the concept of living and user generated services" (European Platform for Intelligent Cities, 2017). There are well-defined pillars of any smart city, such as smart mobility, smart governance, smart environment, and smart people. These pillars help create and enhance smart cities applications such as public health, energy and water services, city management and economic development, transportation, security and emergency services, and waste management (Khan et al., 2013).

The design and operations of smart cities require full integration of ICT-based solutions with the smart cities services. Many technologies can significantly contribute in this regard and UAVs (unmanned aerial vehicles) are one of those that can become an essential component in smart cities (Mohammed et al., 2014a). In this paper we aim to highlight several aspects relating to the integration of UAVs with smart cities components (e.g. infrastructures, buildings, facilities, administrations, and residents). Our study of current and future applications of UAVs in smart cities and their implications, potential advantages and risks is one of the contributions of this paper. In addition, we offer a concise discussion of the technical and non-technical issues that must be addressed to facilitate this integration and lead to an effective, safe

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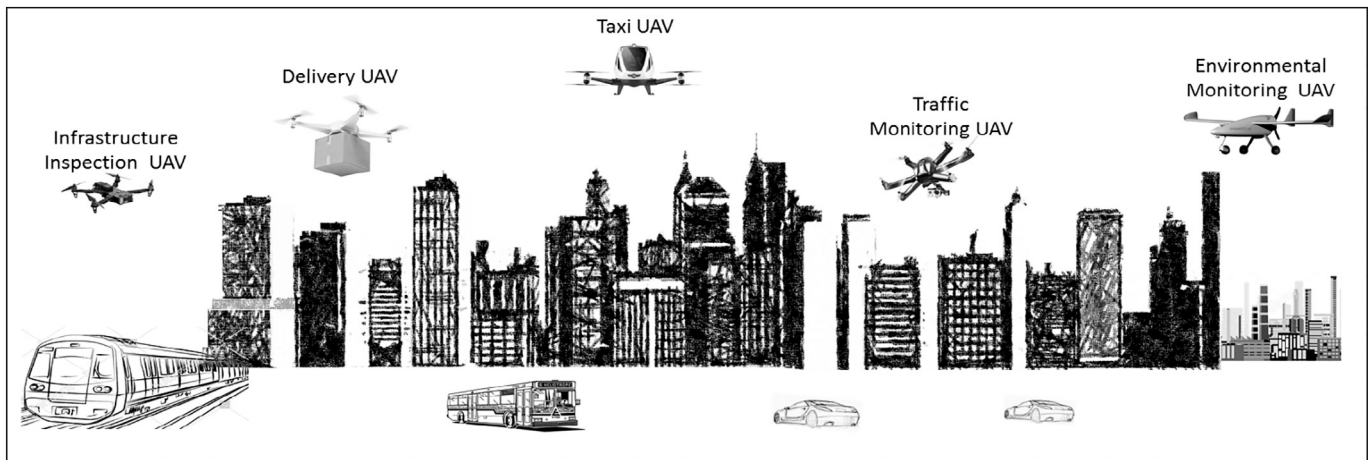


Fig. 1. Utilizing UAVs for different applications in smart cities. UAVs provide the necessary services to achieve the required goals in the smart cities.

and secure set of applications. Moreover, we cover the different enabling technologies (current and future) that will support UAVs and smart cities integration along with the relevant regulations imposed in different countries regarding the use of UAVs.

UAVs are flexible and fast mobile platforms that can be used for many applications in smart cities. These include traffic and crowd monitoring, environmental monitoring, civil security, merchandise delivery, to name a few (see Fig. 1). The UAV applications designed to meet smart cities requirements and functionalities will tie UAVs to smart cities in productive manners. For example, when a UAV is equipped with monitoring sensors and cameras and provided with the software to analyze sensed data from a city's infrastructure, we get applications for smart cities infrastructure inspection and control. UAVs were initially only utilized for military applications. However, they have recently been widely used in civil applications in many domains such as agriculture, logistics, environmental protection, public safety and traffic control. In addition, one of the emerging areas of UAVs is their involvement in smart cities applications resulting in various benefits. Unlike manned planes, UAVs can be more cost-effective. They are more flexible and can operate in various locations and situations, including those that are challenging or posing high risks for humans. They can also fly very close to target objects, which allows for higher measurement accuracy and better targeted actions. These features provide advantages if UAVs are used for smart cities applications. This paper aims to provide insight into the potential applications and issues of integrating UAVs in smart cities.

The rest of the paper is organized as follows: Section 2 provides an overview of smart cities, UAVs, and other related technologies. Section 3 discusses some potential UAV-based smart cities applications. Section 4 addresses the general technical and non-technical issues and concerns of utilizing UAVs for smart cities. Enabling technologies for integrating UAVs in smart cities are discussed in Section 5, while Section 6 discusses some enabling regulations. Section 7 discusses the potential impacts and risks of employing UAVs in smart cities. Section 8 concludes the paper.

2. Background

To set the stage we will first discuss background and overview of smart cities, UAVs and other related technologies.

2.1. Smart cities

The smart cities vision is to enhance the quality of life of the residents and visitors, improve the utilization of the cities resources, increase sustainability, and reduce the negative impact on the

environment. This vision relies on a combination of efficiency and optimization techniques, technological innovations, and historical and real-time data to achieve these objectives. The smartness level of a city can be evaluated by several factors including smart environment, smart living, smart economy, smart people, smart mobility, smart governance, and smart tourism.

Several cities around the world decided to become smart cities for many reasons. These reasons include rapidly increasing populations, limited available unoccupied areas for development, limited resources, increased emphasis on energy efficiency and environmental sustainability, and economic growth. Cities like Malta in the Republic of Malta, Dubai in UAE, Kochi in India, San Jose in California, the United States (US) and several others in Europe already adopted this vision. The European Union dedicated significant investment to develop a long-term strategy for realizing Smart Urban Growth for its cities and metropolitans (Paskaleva, 2009). Dubai is now concentrating on evolving a Smart Dubai initiative that will improve the quality of life for its residents (Mohammed et al., 2014b). It is important with the development of smart cities solutions to consider how smart cities are operated, financed, regulated and planned within the smart cities structure. As a result some structure for smart cities needs to be identified. The authors in (Falconer and Mitchell, 2012) offer a practical structure of smart cities in four layers:

- City indicators.
- City objectives (social, technological, environmental and economic).
- City components.
- City content (solutions and services).

There are several approaches to identify the critical factors influencing smart cities initiatives. As we studied some examples we opted to use the eight core factors offered in (Chourabi et al., 2012) and provide our explanations for them:

- **Management and organization:** the arrangement of management models and organizational goals is an essential requirement for a smart city to work effectively and efficiently to conserve resources and minimize costs.
- **Technology:** a smart city counts on advanced technologies to enhance the monitoring and control of the city's critical infrastructure components and services. Advanced technologies usually involve the new generation of integrated hardware, software and network technologies offering complete solutions with real time awareness and advanced analytics to help make smart decisions.
- **Governance:** it includes the employment of procedures with constituents who provide and exchange information according to

defined regulations and standards to achieve the goals of smart cities. Several aspects like leadership, communication, collaboration, and data-exchange are essential for a good smart cities governance model.

- **Policy-context:** the policy context is an essential factor to the understanding of how to effectively utilize information systems. It helps characterize institutional and non-technical urban issues and creates circumstances to enable the development of smart cities components and services.
- **People and communities:** in addition to improving the quality of life, smart cities initiatives help residents of the cities become active participants in the governance and management of their cities. As a result, they will have an impact on the success or failure of their smart cities.
- **Economy:** it is always considered as a major driver of smart cities initiatives. Applying the smart cities vision usually aims to achieve positive outcomes in encouraging new businesses, increasing new job creation, enhancing workforce development, and improving productivity. In addition, it aims to enhance and optimize the utilization of city resources. All of which will over time positively impact their economy.
- **ICT infrastructure:** the employment of a suitable ICT infrastructure and smart software is an essential requirement to a smart city's success. The performance level of a smart city is highly dependent on the effective use and performance level of the deployed ICT infrastructure and the smart software/hardware services.
- **Natural environment:** one of the core goals of a smart city is to improve sustainability and the natural resources management. This includes the protection of natural resources and associated infrastructures and minimizing the negative impact on the environment.

2.2. Unmanned aerial vehicles

Advancements in robotics include serious enhancements in the design and capabilities of UAVs. Varying sized inexpensive UAVs equipped with microprocessors, data storage, sensors, actuators, cameras, and wireless communication devices are now commercially available. They are reliable and operate with very good level of flight stability. Unlike a few years ago, when UAVs were mainly developed for military applications, numerous civil applications have recently emerged and contributed in further improving UAV capabilities in a more cost-effective manner. The architecture of a typical UAV consists of different components such as the flight system, the control system, the monitoring system, the data processing system, and the landing system. The on-board system usually offers a wide range of functions from navigation, to data gathering and providing data transfers to the ground stations.

When the UAVs were allowed to be used in civil applications, the image of UAVs altered. For example, UAVs were involved in some humanitarian operations where they assist in finding people who need help and to help save lives in many catastrophic situations. They were also used as first responder tools to find out what happened, where the survivors are, how many need help, and what kind of support they need. In addition, UAVs were used for wildlife protection. For example, some guards were trained on how to use UAVs to protect the wildlife in Nepal, which helped stop some major problems. Japan uses UAVs to monitor illegal Japanese whaling in the southern hemisphere. These applications among others provide good impressions about UAVs and encourage their use (Franke, 2013). In addition, UAVs can significantly and effectively contribute in the development of other civil applications such as Surveying activities, environmental monitoring, and traffic control.

Most UAV applications involve mobility and require reaching places that are difficult to access (Ollero et al., 2005). Although, regular manned airplanes or helicopters can be used for these applications, this may involve long periods of repetitive work or operating in dangerous

conditions. It may also require a high level of focus that cannot be handled by some pilots especially for extended periods of time or under stressful conditions. At the same time using regular manned airplanes or helicopters can place a substantial cost and overhead on the department that uses them. Employing UAVs for such repetitive functions typically results in improved productivity lower costs. This is mostly due to the high correctness, flexibility, repeatability levels, and mobility of the UAVs (Saggiani and Teodorani, 2004). Additionally, deploying manned airplanes in risky tasks may put pilots in significant danger; while with UAVs, this risk can be removed or minimized. Generally, the demand on UAVs is rapidly growing. There are many new applications for UAVs being developed continuously. Many public and private organizations are planning to utilize UAVs to enhance the quality, productivity, flexibility, and cost of their services (Kharchenko and Prusov, 2012). However, using UAVs for civil applications creates some issues that we will discuss in this paper.

2.3. Related technologies

In addition to UAVs, the development and operation of smart cities services require utilization of other technologies including wireless sensor networks (WSNs), the Internet of Things (IoT), cloud computing, fog computing, and big data analytics. These technologies are related and essential for many UAV applications discussed in this paper.

WSN is a group of distributed connected sensors used to observe different physical situations such as movement, wind, water flow, temperature, sound, humidity, etc. (Akyildiz et al., 2002). WSNs are used in smart cities to monitor the status of different resources and situations in various areas including monitoring of pollution levels, traffic, buildings occupancy levels, energy and water consumption, etc. (Watteyne and Pister, 2011).

IoT is a network of physical devices such as sensors, actuators and vehicles that enables these devices to communicate and exchange messages. IoT enables the integration of different physical devices within smart cities in a city network (Zanella et al., 2014). Smart cities applications built on this type of network are referred to as IoT smart cities applications. These applications use the IoT components and other needed systems to perform their operations.

As defined by the National Institute of Standards and Technology (NIST), cloud computing is a paradigm allowing users to access shared configurable computing resources through standard protocols (Badger et al., 2011). One of the main features of cloud computing is that computing capabilities can be quickly and easily scaled in or out based on demand. Cloud computing can provide different computation, storage, and advanced services for smart cities applications. Examples of advanced services include data mining, machine learning, simulation, optimization, and autonomous decision making (Mohamed et al., 2017d).

Fog computing is an architecture that uses one or more edge devices like advanced network routers, computers, or other compute-capable devices available near the IoT applications to provide localized services for these applications. These services can be storage, security, communication, control, configuration, monitoring, management or mobility support services. Using fog computing will provide better support for smart cities IoT applications requirements such as low latency, location awareness, mobility, streaming and real-time responses (Giordano et al., 2016). Fog nodes are distributed in various locations within a smart city to effectively provide their services.

Big Data analytics is the process of examining, cleaning, converting, and modeling exponentially growing data with the aim of realizing beneficial information that improve different decision-making processes (LaValle et al., 2011). Big Data Analytics can be used to facilitate smart (may also be autonomous) decision making based on the collected data to enhance smart cities services (Al Nuaimi et al., 2015).

3. UAV applications in smart cities

This section discusses several different current and foreseen UAV applications for smart cities. These applications provide beneficial advantages for smart cities to improve services performance and residents quality of life.

3.1. Traffic monitoring and management

One of the major problems facing many large cities is traffic congestion. This problem can occur due to periodic or sudden increases of the number of vehicles for reasons like rush hours, large events, construction work, or accidents. These problems happen any time and at any place within these cities. This forces transportation administrators to search for more effective and efficient ways to decrease traffic congestion. To arrive at an adequate solution, we first need to know the congestion areas, causes, vehicles volume, and the status of the roads around the area. Although static cameras installed on a city's streets can provide some information about these factors, they usually do not provide the complete real-time information needed to manage the problem (Barmounakis et al., 2016). UAVs can be used to collect and deliver a complete set of real-time information about traffic congestions. As they have the mobility and high visibility advantages, UAVs can provide live feeds covering the congestion area and all surrounding areas. Several approaches were developed that offer different types of effective and efficient methods of communicating and exploring traffic data collected from UAVs (Bethke et al., 2007; Chen et al., 2007; Kanistras et al., 2015; Kim, 2005). Some approaches also help study the types of traffic data that should be gathered and the suitable filtering processes by the UAVs. In addition, there are some efforts to develop traffic simulation models that utilize collected traffic data to help evaluate different possible solutions to solve congestion problems (Coifman et al., 2006; McCormack and Trepanier, 2008; Heintz et al., 2007).

Another reason that can contribute to traffic congestion in some areas in large cities is when multiple cars are trying to find parking spots. With events happening and limited parking spaces, drivers tend to circle the target area several times, usually at low speeds, in hopes of finding a spot to park their vehicles. A solution was proposed to try to find empty parking spots using UAVs (D'Aloia et al., 2015). This solution can be integrated with other automated systems to help drivers find or reserve unoccupied parking spots faster. In this context, UAVs can become a valuable contributor to the traffic management efforts. In addition, incorporating more data sources and monitoring and data collection techniques will lead to even better and more accurate solutions. However, this will also raise security and privacy concerns. An example could be the possibility of causing major problems by hacking into these systems, or illegally exploiting private data about the commuters and vehicles.

3.2. Health emergency services

Accessibility to basic life support and emergency healthcare services anywhere and at any time is an important issue that every city needs to make available to its residents. UAVs can be used to provide or support health emergency services. They can be used to quickly deliver needed medical supplies and services to patients. Such services will offer prospects for basic life and health support systems to be available rapidly to patients. As an example, a UAV can be used to transport a defibrillator and the needed medical devices directly to a cardiovascular patient to offer cardiopulmonary resuscitation (RT-Russia, 2014). Another example is UAVs that could operate as ambulances for remote or congested places. Using UAVs for these situations to provide ambulatory services and deliver essential life support equipment and medical supplies can speed up responses and cut the costs of these services to for inaccessible areas. UAVs can also offer similar emergency services to

people in public facilities when accidents occur within the city that limit or cut off ground transportation. Man are investigating was to enhance payload capabilities, power efficiency, and the ability to handle difficult weather conditions to improve the emergency responses using UAVs. In addition, it is also important to determine the limits at which such UAVs can operate without compromising public safety or leading to legal issues or unintentional injuries.

3.3. Security and crowd monitorings

UAVs can assist policemen to enforce safety and security measures in smart cities. They can be deployed quickly to collect real-time information and support these operations in various situations. For example, the United Kingdom policemen used a UAV to help in arresting a car thief suspect (Hull, 2010). Generally, managing the effectiveness and efficiency of security systems and operations in smart cities has become challenging. Using UAVs for security monitoring and support can provide some advantages such as offering more accurate and real-time information for authorities to make the right decisions and enforce suitable security actions. Utilizing UAVs for policing large events in cities have lately been supported by the US congress and top level federal agencies such as the US Department of Justice and the Bureau of Justice Assistance. Integration of UAVs with other technologies like video streaming, secure and reliable wireless communications, forensic mapping software, video-based abnormal motion detection (Kiryati et al., 2008) and video-based abnormal human behavior recognition (Popoola and Wang, 2012) can assist in making smart cities a safe place for living. UAVs can be used to monitor crowds participating in large events such as major outdoor celebrations, sports games, and parades. It is imperative for security enforcement and safety concerns in such events to monitor crowd movements in real-time. UAVs can be utilized for surveillance, which improves accuracy and reduce costs, thus enhancing the productivity and efficiency of the surveillance operations. This also allows for better information gathering and quicker decision making. The information can be used to increase police presence in some places or close some streets to smooth pedestrian movement. In addition, they will provide valuable input in case of disturbances or catastrophic incidents enabling the authorities to provide the needed ground operations, crowd control and medical services where needed most. UAVs can also be used to enable efficient information flow for the current situation.

3.4. UAV-based infrastructure inspections

UAVs can provide flexible, fast, and cost-effective tools for inspecting city infrastructures (Mota et al., 2014). While routine inspections for some city infrastructures can be good to avoid major problems that might cause human deaths or/and can be very costly to recover from, these routine processes usually involve huge costs and efforts. UAVs can be used effectively in some of these inspections. In addition, UAVs can be sent rapidly anytime to check in more details on reported problems in these infrastructures. UAVs can provide images and scans of possible problem areas. These images and scans can be analyzed with software to find indications for any problems in the infrastructures. Many proposals were developed to utilize UAVs for inspections for bridges (Metni and Hamel, 2007), power lines (Deng et al., 2014), pipelines (Hausmann et al., 2003), and large buildings (Hallermann and Morgenthal, 2014).

3.5. Agriculture management and environmental monitoring

UAVs can be utilized to help support agricultural processes like distributing fertilizers, pesticides, seeds, and water. They can also provide periodic inspections of crops and their conditions. Furthermore, they can be utilized to screen the development of crops to decide on best harvesting times or if some protective measures are necessary. For

examples, some crops can be heavily damaged with sudden changes in weather conditions. Grapes can be damaged when frost sets in and farmers need to quickly provide some warmth through controlled fires or special covers. UAVs can help detect frost conditions early and become an aeration component to spread warm air using their rotors or specially designed fans onboard. In addition, piloted planes were used to spread pesticides and other necessary components on fields, yet UAVs are very capable of taking responsibility of these tasks.

In another relevant topic, UAVs can also help in monitoring and measuring environmental conditions such as measuring for harmful substances like high CO₂ levels and air pollutant presence. This allows for early detection of harmful conditions and accurate assessment of the risks involved. Thus enabling early intervention when necessary. In some sense, we find this type of UAV applications in smart cities as one of the least controversial type. Issues like privacy and confidentiality are minimal and human related risks are also very low. A main risk would probably be hacking the UAVs system for commercial espionage or to sabotage the farms.

3.6. Tourism support

UAVs can be used as guides for tourist attractions such as historical and natural museums, zoos, and other cities' facilities. It can direct visitors around the attraction area and connect and provide information to them using smartphone apps (SKYCALL, 2013) or onboard audio systems. The used UAVs can be equipped with onboard cameras with different functions. They can be used as monitoring cameras that relay images to the main office. They can also be used as cameras that can be controlled and accessed by the visitors to take personalized and customized photos. UAVs can be used to guide visitors to specific sceneries they would like to see. Moreover, UAVs can provide multi-language translation capabilities based on visitors' needs. In addition, tourists in the future may be able to ride special UAVs to take them autonomously in areal tours and sightseeing. Safety and security issues are the main concerns in the latter applications.

3.7. UAV-based surveying

Surveying has many applications in civil engineering and in city management. However, most traditional surveying methods consume huge efforts and time. They can be very costly, inaccurate, and may involve dangerous activities for the workers. With advances in UAVs, remote sensing, and image processing technologies; UAVs can be used effectively in geospatial and surveying activities in smart cities. UAVs can function as flexible, fast, accurate, inexpensive, and safe tools to conduct different surveying activities needed for construction projects or city management systems. A UAV can be used to acquire three-dimensional (3D) mapping data for large infrastructure projects (Siebert and Teizer, 2014). UAV-based surveying can generate different types of 3D mapping such as textured 3D models, digital surface models, terrain surface models, vector information, and contours (Nex and Remondino, 2014). In addition, UAVs can help in performing geospatial land surveying of different parts of a smart city. Acquired data can also be fed to Geographic Information Systems (GIS) for different types of smart cities management applications.

3.8. Large-scale disaster management

Situation and emergency response management in large-scale disasters such as earthquakes, floods, volcanos, fires in forests or large infrastructures, and terrorist attacks, are very challenging. In such disaster circumstances, fast efforts in situation monitoring, analysis, emergency response arrangements, and control are extremely important. However, in most of these circumstances humans and emergency teams cannot easily and quickly reach the disaster areas or each other (in some cases). At the same time, most of the infrastructures such

as roads and telecommunication systems may be damaged. This makes the regular monitoring and emergency response actions harder to achieve effectively. UAVs can be utilized effectively in such circumstances (Ataei and Paschalidis, 2015). They can be used as flexible, reliable, and safe tools to monitor and provide real-time information about the current situation. For example, two major natural disasters occurred in Fukushima, Japan in 2011. The first was an earthquake of magnitude 9.0 (recorded as the strongest earthquake in Japanese history). The second was a major 15-metre tsunami which damaged the whole area, initiated a nuclear disaster at the Fukushima Daiichi Nuclear Power Plant, and killed thousands of residents. There were no other than UAVs to monitor the situation of these disasters (Madrigal, 2011). UAVs can be allocated to different places to get more real-time information for better analysis and assessment of the situation. UAVs can also be used to perform some actions to control the situation in some disasters. For example, they can be used as fire fighters for risky industrial facilities and large commercial and residential buildings (Estes, 2014). In addition, UAVs can be utilized to launch alternative communication systems to substitute the damaged ones for emergency use (Tuna et al., 2014). Furthermore, UAVs can be used to help locating survivors, transporting medical supplies and equipment, and may even function as ambulances to transfer the injured.

3.9. Merchandise order delivery

UAVs can be used to deliver customer orders within a short time. As a customer places an order online, he/she can receive the order quickly through a delivery UAV designed for this specific application. UAVs are flexible to move fast to different areas in cities for rapid delivery even in crowded cities. Using the airspace basically increases mobility by avoiding congestion areas and utilizing faster routes. Companies like Amazon and DHL started developing and testing this concept (Heutger and Kückelhaus, 2014). In addition, Google is developing a system named Project Wing to construct autonomous delivery systems to deliver packages to different places within a few minutes. Google conducted preliminary testing for rural deliveries in Queensland, Australia (Nieva and Rosenblatt, 2014). With such applications taking off, we may have some issues with air traffic control and security.

3.10. UAV-aided wireless communications

The availability of reliable and scalable wireless communication is an essential factor in providing many successful interactive services in smart cities (Jawhar et al., 2017). For example, advanced smart transportation services cannot be provided without a reliable and scalable wireless communication system linking the different entities involved in this application. Wireless communication in smart cities can be provided and utilized for different applications using different communication technologies such as Wi-Fi, GSM, WiMAX, and satellite links. However, due to some issues some of these technologies may not be able to continuously provide reliable wireless communication services. For example, due to natural disasters, some base stations may get damaged and become unusable. Another example is when some base stations get overloaded in extremely crowded areas because of special events such as sport games. In such cases, UAVs equipped with wireless communication support devices can be deployed in these areas to provide basic or additional coverage for different clients of smart cities services (Zeng et al., 2016). In addition, UAVs can be used to assist in offering wireless connectivity between distant areas to continue providing interactive smart cities services.

3.11. UAV taxi

Recent development efforts are conducted to use UAVs as air taxis in smart cities, bypassing traffic congestion. One of the main goals is to provide a safe, autonomous, and unmanned UAV taxi on demand.

Another goal is to offer faster transport by utilizing more direct travel routes and avoiding the traffic congestion of large cities. Dubai recently performed its first experiment of a UAV taxi service as part of its smart city vision (Wakefield, 2017). This experimental UAV taxi can carry two passengers and is developed by a German-based company named Volocopter. This company aims to have their UAV taxis in operation within 5 years. Although, there are good advantages for this application, there are specific challenges that need to be fully resolved and tested. These challenges are represented in two points. The first is that a UAV Taxi that will carry passengers must have extremely high safety and security requirements to protect the passengers. These requirements may significantly increase the cost of providing such a service. This takes us to the second point which is the cost-effectiveness of providing this service. UAVs designed for unmanned operations can disregard several of the safety regulations; however, when humans are the payload, these regulations must be strictly applied, thus significantly increasing the cost of building and operating such UAVs. Other issues will arise when this concept actually becomes available and cost effective leading to an increasing number of UAV taxis in the air. Air congestion, landing pads use, air traffic control and legal and ethical issues will arise.

3.12. UAV-based fog computing

Many smart cities applications are based on the IoT (Zanella et al., 2014) using sensors, actuators, and other wirelessly connected devices. Some of these applications need to be integrated with the cloud and other systems for various purposes. One example is utilizing advanced services provided by the cloud such as elastic resources at reduced costs. IoT applications can utilize the cloud for data storage, powerful processing, and advanced services. Nevertheless, integrating IoT applications with the cloud has many restrictions as the cloud cannot deal with the essential characteristics and requirements of IoT applications such as highly heterogeneous devices, mobility, low-latency responses, and location and context awareness. To overcome these restrictions, the concept of fog computing was proposed by Cisco (Bonomi et al., 2012). Fog computing can improve the cloud computing paradigm by offering smaller platforms located at the network edges closer to the IoT devices and networks. Using Fog Computing, an application in a certain area can utilize an architecture that uses a dedicated computer available locally, or one or more clients on nearby edge devices. This provides access to computing and storage services at a smaller scale, but very close to the application, thus reducing response times and providing localized services. As a result, access to cloud services can be minimized, yet efficiently accessible when needed.

Fog computing is proposed to enhance the operational efficiency for many smart cities applications such as smart transportation, energy management for smart grids and smart buildings, and city infrastructure monitoring (Mohamed et al., 2017c; Perera et al., 2017; Yannuzzi et al., 2017). Fog units can be distributed through a smart city to provide support for different smart services. A new paradigm of UAV-based fog computing was introduced recently (Mohamed et al., 2017a). This paradigm combines the advantages of fog computing and UAVs to better support IoT applications by utilizing UAVs equipped with fog units. UAV-based fog computing can provide flexibility, mobility, and fast deployment features to support IoT applications in a smart city. UAV-based fog can be used for different purposes. One is to rapidly (and temporarily) replace a faulty fixed fog unit. An example is to dispatch a UAV-based fog to replace a faulty fixed fog unit that provides intelligent controls for traffic lights. Another purpose is to temporarily deploy a UAV-based fog in an IoT application area to add more capabilities to a currently loaded fixed fog. In addition, in disaster situations, UAV-based fog units can be deployed to support search and rescue operations in place of lost ground fog units.

3.13. Data collection in WSN using UAVs

The wireless sensor network (WSN) technology has recently improved and is being used for many applications. Some sensor network applications require communication between the distributed nodes and control stations to transfer observations in real-time through multi-hop communication. Others are delay-tolerant and do not require to transfer their observations in real-time. Examples of applications that do not necessarily need real-time communication are infrastructure health monitoring and environmental monitoring, where data processing needs significant periods of time. In these cases, UAVs can be used as data collectors from these distributed sensors to transfer them to the main stations (De Freitas et al., 2010; Jawhar et al., 2013). The UAV can pass over these nodes and wirelessly collect their data. This approach can save the sensor nodes' energy by reducing its communication needs and extend the life of the WSN (Jawhar et al., 2014). There are several strategies developed for efficient data collection in such applications (Jawhar et al., 2015, 2016; Oliveira et al., 2010). In addition, there are some efforts to enhance communication protocols between UAVs and sensor nodes (Ho and Park, 2010; Ho and Shimamoto, 2011; Ho et al., 2011).

4. UAVs utilization issues

Although UAVs can offer many benefits for smart cities services, effective utilization of UAVs poses several issues that need to be addressed. These can be classified into technical and non-technical issues.

4.1. Technical issues

The technical issues are those involving the hardware, software, communication and operations of UAV systems.

- a. Safety: One of the most important issues of utilizing UAVs in smart cities is safety. UAVs in smart cities will operate among buildings, roads, and other city infrastructures. They will also operate in some cases very close to people. As a result, successful integration of UAVs into smart cities requires incorporating high safety requirements. This requires designing UAVs in ways that ensure safe operations for city residents, buildings and infrastructures. Regulations concerning proximity, speed, altitudes, and maneuverability must be put in place, enforced and adhered to by UAVs. Although, sense and avoid approaches for UAVs have improved significantly (Angelov, 2012; Carnie et al., 2006; Korn and Edinger, 2008; Ramasamy et al., 2014; Shakernia et al., 2005), these approaches need to be tested and evaluated in urban settings.
- b. Security: Access and control of UAVs and their services for smart cities should only be done by authorized entities (Vattapparamban et al., 2016). No unauthorized entities (people, organizations, or other systems) should gain any type of access to the UAVs. In addition, UAVs controlled by authorized entities must not fly over unauthorized locations within the city. The communication between UAVs and Ground Control Stations (GCS) should also be protected. Furthermore, all control subsystems within the UAVs and GCS should be protected. Generally, using UAV systems in smart cities require providing protection against several types of attacks (Rodday et al., 2016). One type is malicious hardware attacks done through intentional tampering with UAVs hardware to alter their designed behaviors. Another is flight control computer attacks in which some external entity interferes with the UAVs internal flight control and communication systems to change their actions or mission configuration parameters. Another type is navigation sensors attacks which can alter a UAV's planned path. Here, a UAV could operate as if it is at its correct location while in reality it has been led to a different location. Other attacks are communication channels attacks and GCS attacks. Channels among UAVs or UAVs

- and GCS are vulnerable to attacks especially if the communication signals are not encrypted. The computers running the control software and communication systems in the GCS also need to be protected. One of the good approaches proposed to reduce such threats is to use a UAV security monitoring approach (Birnbaum et al., 2015). With this approach unplanned flight and action deviations due to some attacks can be detected. While different aspects of securing UAV applications for smart cities need to be developed, it is possible to benefit from other security and privacy techniques developed for other fields such as VANET (Vijayakumar et al., 2018) and IoT (Liao et al., 2017) including different security and privacy mechanisms.
- c. Reliability: UAVs used for smart cities should be reliable. UAVs may encounter hardware failures like mechanical or electronic components failures or malfunctions which may restrict the ability of UAVs to complete their tasks. In a smart city, UAV malfunctions or damages and the impact of such failures must be minimized. One possible and well-known approach to solve this issue is to use adaptive control algorithms with hardware failure detection mechanisms (Zachary et al., 2010). These algorithms allow for adding fault tolerance features to handle some types of failures and avoid their impact. Another aspect to consider is the failures in communication, which could lead to different types of problems ranging from simple delays in delivering data to catastrophic losses of important control or mission instructions. Several approaches are available to ensure minimized losses in communication including deploying redundant communication channels, using enhanced multi-channel communication and deploying strong transmission controls. Software failures are also a possibility, however, most of these can be detected and corrected before actual deployment through rigorous testing and systematic verification and validation of the software developed for the UAV systems.
 - d. Communication: UAVs used for smart cities generally require communication links with GCS and other systems in smart cities. In addition, some UAV applications such as collaborative UAV applications require communication links among the operating UAVs (Tortonesi et al., 2012). These applications usually dictate specific communication requirements and QoS levels in terms of bandwidth, delay, transfer rate and jitter. Most UAV applications for smart cities cannot maintain safe, secure, and reliable operations without these requirements. There are different communication architectures that can be used such as satellite, direct link, cellular, and mesh networks (Frew and Brown, 2008), which have different characteristics. A careful selection of one or more of these architectures is needed to best support the communication requirements of the deployed UAV applications. For example, direct links between the UAVs and other systems in a smart city may be the simplest technique to use. However, obstructions such as tall buildings and mountains can negatively impact the communication signals. Also, longer communication ranges require high-power transmitters and special antennas to achieve direct communications. These requirements make the direct link unsuitable for small and limited resources UAVs. In addition, the available communication bandwidth may be influenced by the number of UAVs used in the area. At the same time, high communication delays will occur if multiple collaborative UAVs operating in the same area communicate through satellite link. The cellular architecture provides several advantages; however, it has also several limitations as it is not designed to support UAVs. For example, UAVs can easily degrade the performance of cellular networks due to their high communication demands compared to cellular phones. A possible solution is using a dedicated cellular infrastructure for UAVs in smart cities. The mesh architecture, where all UAVs and GCS are used as relay nodes to forward data, is an attractive architecture for solving many challenges for UAV communication. However, it always needs high security requirements to be effectively used for UAV applications (Finn and Scheduling, 2010; Liu et al., 2004).
 - e. Cost Effectiveness: As different types of services in smart cities will be provided or supported using a large number of UAVs, these systems should be designed and developed in an economically efficient manner in terms of the cost of both development and operations. The development costs can be reduced by producing more UAVs for civil applications (economy of scale). In addition, using modularity and suitable software architectures can reduce software development costs. With modular designs, common functions and components can be used for different UAVs with varying functions, thus reusability can be enhanced and the development costs can be reduced (Ogren, 2012; Tisdale et al., 2006). Furthermore, using advanced levels of UAV autonomous controls can reduce the need for manual controls and human involvement thus the operations costs can be significantly reduced (Chen et al., 2009).
 - f. Heterogeneity: Different applications utilizing UAVs in smart cities require the use of heterogeneous UAVs capable of efficiently performing different tasks. These UAVs may have different payload specifications and components, varying operational boundaries, and mission-specific operational characteristics. Therefore, developing and operating these heterogeneous UAVs within a single system can be challenging. Abstraction and modular design can help reduce some of the complexities of dealing with heterogeneity. Moreover, careful design of interfaces across different UAVs, their payloads, and the software components used will further improve the situation.
 - g. Integration with other Systems: UAVs have to be seamlessly integrated with other systems to operate effectively within smart cities. These systems can be control and monitoring applications, enterprise systems, or cloud/fog services. This integration is needed to support information exchange between different systems and UAVs. Integration with other systems can be challenging in some cases as systems differ in their operational parameters and capabilities in addition to having different interface models. Some level of agreement on interfaces and access methods can help. Particularly, adopting an advanced middleware framework can facilitate the integration across different systems (Mohamed et al., 2013).
 - h. Development and Testing Environments: Testing and validating UAV applications for smart cities is extremely difficult as it is not feasible to physically run testing experiments in the smart cities. Therefore, most of the tests are done within the limited development and testing environment that cannot fully mimic the real environment or using computer-based simulation models, which also cannot represent every aspect of real life situations. So far, no realistic development and testing environments exist that can closely provide real and to scale smart city settings to more accurately test UAV applications. Although there are some general efforts in that regard, there is no sophisticated and comprehensive environment that include the development tools and testing models needed to cover the challenges of smart cities.
- #### 4.2. Non-technical issues
- There are some non-technical issues affecting the use of UAVs for smart cities applications. These involve various factors that may raise questions about the safe and appropriate use of UAVs within smart cities.
- a. Privacy: UAVs are usually equipped with several sensors and cameras to collect data. This may raise privacy concerns for residents and some organizations in the cities where UAVs operate. Although specific goals are given to UAVs operating within a city, some may not be happy to have them collecting data in or around areas they are using. Having regulations and clear guidelines regarding this issue is essential to ensure proper use of UAVs and peoples'

acceptance of such use. In some countries like the United Kingdom, residents are protected by a Data Protection Act which needs to be further developed to include UAV use. Different privacy issues of using UAVs and some solution directions and recommendations are discussed in [Cavoukian \(2012\)](#).

- b. **Ethics:** UAVs in smart cities may be used for different applications and under the control of multiple public or private organizations. Therefore, there are possibilities that UAVs are misused for purposes other than their originally intended operations. Examples include using the UAVs to spy on residents or organizations, or using data collected by the UAVs to influence certain persons or organizations when making decisions. This can represent an obstacle for using UAV applications in smart cities ([Finn and Wright, 2012](#)). There is a need to develop and apply a code of ethics for utilizing UAVs in smart cities. This code of ethics should professionally define guidelines and solutions for different ethical issues that may arise with using UAVs in smart cities. Policies and procedures for proper and ethical use of UAVs must be established and lawfully enforced.
- c. **Cost:** Developing UAV projects can be expensive because of their technical and deployment issues and the training and integration of systems costs. Designing a UAV system for a specific service is also expensive as it needs to be carefully designed, implemented and tested to function properly. The total cost can increase even higher for UAVs that will carry people as in the UAV Taxi and UAV ambulance as the safety and comfort requirements will be very high and complex. ([Malone et al., 2013](#)) described a cost estimation requirement for developing UAVs and a framework for estimating a UAV system's total ownership cost. It includes hardware components, software design, and operations. Generally, the type of UAV application defines the required speed, altitude, endurance, and payload all of which determine the total cost requirements.
- d. **Licensing and legislations:** UAVs, depending on their size and mission, will operate in different areas of airspace. Some may interfere with other systems like commercial flight routes or helicopter space. To use UAVs in any country it should be first officially registered. In addition, UAVs must be cleared to use specific altitudes and areas. Therefore, smart cities governing bodies must develop regulations and legislations for UAV deployment and use in their airspace and define specific procedures for enforcing them. UAV operators must then adhere to these regulations for safe operations. Another aspect involved is the legal issues relevant to the actual use of the UAVs and their impact on people and infrastructures, where many challenges may arise and must be addressed by the legal systems in smart cities. As a result, the legal system must be augmented by laws and rules pertaining specifically to UAV applications and their effects.
- e. **Business adoption:** It is a challenge for companies to introduce UAVs in some aspects of their business due to the costly additional resources and skills involved. However, if UAVs are used, they can be very helpful for the business to gain strategic advantages. Businesses can utilize UAVs to provide services with some qualities that cannot be offered by their competitors. For example after Amazon introduced their UAV for deliveries, it created a huge competition with UPS, which is another delivery giant. Prime Air UAVs can deliver products purchased from [Amazon.com](#) to the customers within 30 min, which can be very attractive for some customers. Businesses and organizations need to weigh in the benefits of such applications versus their costs and implications and decide on a suitable approach. However, as more UAVs enter the market and many types and even software components become mainstream leading to mass production, some of these costs should start to drop making it more feasible to adopt.

5. Enabling technologies

Developing UAV applications require the use and adoption of

various supporting technologies. Some technologies have matured and can be easily adopted, while others are still in their early stages, yet offer promising capabilities to support efficient and effective UAV applications. Here we discuss four main technologies that will impact the development and utilization of UAV applications for smart cities significantly.

5.1. Autonomous UAVs

Most of the traditional UAVs used for civil applications are restricted to Visual Line of Sight (VLOS) operations. In this type of operations, the pilot must be able to see the UAV while in operation. The control can be through joysticks; computer-based like smartphones, tablets, or laptops; or gesture-based interfaces ([Mashood et al., 2015](#)). However, most applications in smart cities need UAVs to operate autonomously beyond the VLOS (BVLOS) and with minimum human interaction. Fortunately, good progress has been made recently in UAV technologies that provide effective BVLOS flight operations. UAVs that can provide BVLOS flight operations should have several characteristics and capabilities including safety and reliability, collision avoidance, broadband communication, and real-time tracking. While civil aviation authorities have not yet given permissions for BVLOS operations, there are releases given to UAV manufacturers to experiment with BVLOS flight operations ([Hini, 2017](#)). Successful production and testing of UAVs with BVLOS operations will enable many applications in smart cities such as traffic tracking, environmental monitoring, and merchandise delivery.

5.2. Integration technologies

The infrastructure of smart cities usually incorporates many heterogeneous systems with varying software and hardware components. These can be sensor networks, servers, cloud computing, fog computing, smart transportation systems, to name a few. Consequently, smooth and seamless integration of UAVs into smart cities systems is not trivial. There is a pressing need for innovative middleware frameworks to facilitate integration. One example of such advanced middleware platforms is using service-oriented middleware (SOM) ([Al-Jaroodi and Mohamed, 2012](#)). SOM can view UAVs' functions as a set of services. It can also represent various types of devices and components in the smart cities as services. All these services can be provided using standard interfaces such as web service interfaces. In addition, they can be registered, looked up, and consumed by other services or systems to build complete UAV solutions.

Another aspect of integration arises when smart cities applications require multiple collaborative UAVs to complete the target tasks faster and with more accuracy and reliability. Examples of these UAV applications are national disaster monitoring, traffic monitoring, and WSN data collection. Here, there is a need to use a middleware platform for collaborative UAVs ([Mohamed et al., 2013, 2014](#)). Such middleware can facilitate building and operating collaborative UAV applications by providing specially designed services for collaborative sensing and acting, collaborative data processing, collaborative communication, and collaborative control between UAVs and other smart cities systems. Another approach to enable collaborative UAV applications is to use cloud computing ([Mahmoud and Mohamed, 2014](#)). The cloud can provide a broker for UAVs resources and services that can be used for several UAVs and applications ([Mahmoud and Mohamed, 2015](#)). The Integration of UAVs into the Cloud can be achieved using the concept of the web of things ([Mahmoud et al., 2015](#)), which also enables the concept of UAV as a service ([Koubâa et al., 2017; Yapp et al., 2016](#)).

Middleware can also be considered as an approach to provide advanced services for UAV applications in smart cities. One example is providing adaptive data processing and dissemination for UAVs in urban sensing ([Wu et al., 2017](#)). In this advanced service, online learning techniques are used to adaptively balance the data processing

Table 1
FAA airspace classification.

Class	Height range (feet)	Usage
A	18,000–60,000	Commercial airlines fly range
B	10,000	Big commercial airport airspace
C	4000 and above	Small commercial airport airspace
D	2500 ft and above	Small commercial airport airspace
G	Controlled airspace 700–1200	UAVs in civil applications

and dissemination processes. In addition to this advanced middleware service, other advanced services closely tailored to the needs of smart cities UAV applications can be developed and deployed. Generally, using an advanced middleware and other integration technologies for smart cities UAV applications will significantly decrease the development, testing, and deployment costs of such applications.

5.3. Cloud and fog computing

Integrating UAVs to cloud and fog computing platforms can enable many applications by providing advanced services for utilizing UAVs in smart cities. Both the cloud and fog can provide powerful services for the operations of the UAVs. For example, using a cloud-based framework for risk-aware intelligent navigation in cities (Primatesta et al., 2017). In this framework, cloud-based control, risk-aware navigation, and planning algorithms are used in real-time. Furthermore, a cloud-based system for unmanned air traffic management within cities was proposed in (Foina et al., 2015). This system provides traffic control and management services to enable safety and collision avoidance.

Another example shows the benefits of using fog computing. Many UAV monitoring applications involve using cameras and remote sensors to collect and analyze huge amounts of data in real-time. This data cannot be analyzed using the limited capacity microprocessors onboard the UAVs. Yet it is not realistic to rely on the cloud to do the processing due to the limitations in communication bandwidths and high levels of delays. Fog nodes can be used in this case to augment the need by making available the necessary processing power closer to the operational locations of the UAVs. For example, fog computing was used for dynamic urban surveillance video stream processing (Chen et al., 2016). Generally, fog computing can provide good real-time support for moving camera analytics (Lin et al., 2015).

While both cloud and fog computing can provide many advantages to support UAVs operations and applications, the integration of the cloud and fog computing with UAVs can be an issue. As discussed in the integration technologies needed, middleware can play a significant role in supporting seamless and efficient integration of smart cities UAV applications with cloud and fog services. SmartCityWare, a service-oriented middleware, was developed to provide support for such integration (Mohamed et al., 2017b).

5.4. Big data analytics

Smart cities services can utilize big data analytics for intelligent decision-making processes (Moreno et al., 2017). Utilizing UAVs for smart cities will increase the collected data to analyze. This increase will provision better decision-making and optimizations for different services provided by the smart cities. Combining big data analytics with UAVs as mobile platforms to collect data becomes essential to help enhance services in smart cities. For example, both UAVs and big data analytics can be utilized to enhance disaster response efforts (Ofli et al., 2016). With the availability of powerful infrastructures such as cloud computing, big data analytics can be effectively incorporated in many applications including smart cities UAV applications. For example, utilizing the cloud and big data analytics for the UAV disaster response application will provide vital, accurate and fast responses to rescue

people, control the situation and minimize damages (Chang, 2017).

6. Regulations

Using UAVs for smart cities may require them to operate within the airspace utilized by the regular manned planes used for military, police, cargo, and passengers. As a result, UAVs cannot safely operate within any city without having specific use regulations defined and properly enforced. Currently some governing bodies have put through regulations involving airspace use, communications and licensing and certification. These regulations create and enforce some level of control on UAV use, however, more effort is needed to refine current rules and extend them for the foreseen wider range and heavier use of smart cities UAV applications.

6.1. Airspace for UAVs

The United States Federal Aviation Administration (FAA) (FAA, 2017) that authorizes the civil utilization of unmanned aircrafts, defined and enforced several regulations. UAVs utilized for civil applications are constrained to fly within specific altitudes. Table 1 lists the airspace classes based on the FAA classification. Class G is the airspace allocated for UAVs used for civil applications while classes A, B, C, D, and E are allocated for manned commercial aircrafts. FAA regulated class G airspace, between 700 and 1200 ft and categorized UAV operations in three groups: Civil, Public and Model Aircrafts. These are regulated by section 333 of the FAA Modernization and Reform Act of 2012, “Special Rules for Certain Unmanned Aircraft Systems,” which allows UAV commercial operations in low risk controlled airspace (FAA, 2014c). This restriction offers a safe operating environment for the UAVs, yet at the same time limits their operational area. As UAV applications increase, issues of congestion and traffic control will start to arise and more rules and regulations will need to be put in place.

6.2. Communication for UAVs

Using UAVs within cities also requires defining new communication solutions which can effectively support UAV operations within the cities' airspace. NASA (National Aeronautics and Space Administration) in the US and EUROCONTROL (European Organization for the Safety of Air Navigation) in Europe are managing the development of new communication systems for enabling the operations of UAVs. The US is developing and testing a standard for UAV Control and Non-Payload Communication (CNPC) links to enable safe integration of UAV operations within the National Airspace System (NAS) (Kerczewski and Griner, 2012). NAS is the main aviation system in the US that includes, US airspace, airports, and monitoring and control equipment and systems that offer the enforced regulations, policies, and procedures. This system covers the airspace over the US and large portions of the oceans and some of its components are shared with the air force. For safe incorporation of UAV operations in NAS, several policies, requirements, and procedures are being defined. These include aircraft-human interfaces, sense and avoid techniques, air traffic management policies and procedures, certification requirements in addition to the CNPC development (NASA, 2012).

CNPC links are developed and defined to offer a set of communication connections for different purposes. These connections are used to support UAVs safety functions and to allow for control and monitoring of UAVs by ground stations and remote pilots. This involves several issues including communication architecture types, rate and bandwidth requirements, frequency spectrum allocation, and secure and reliable communication provisions. Two architecture types were proposed including line-of-sight (LOS) communication, which provides communication with unmanned aircrafts through ground-based communication stations and beyond-line-of-sight (BLOS) communication, which provides UAV communication through satellite links. The

Table 2
CNPC data communication rates.

Aircraft size	Uplink rate (bps)	Downlink rate (bps) (usage)
Small (≤ 55 kg)	2424	4008 (Basic service only)
Medium & large (> 55 kg)	6925	13,573 (Basic service only) 34,133 (Basic service & weather radar) 234,134 (Basic service, weather radar, & video)

communication rates requirements for both uplink (ground-to-air) and downlink (air-to-ground) were defined based on the size of the aircraft as shown in Table 2. The uplink rates are much lower than downlink rates as the uplink communication will be mainly used to send small messages to control unmanned aircrafts while the downlink is used for different types of communications including data and video transmission. The uplink rate was determined to support transmission of 20 individual control messages per second. This rate is required to provide a complete real-time ground control for a UAV using a joystick (Kerczewski and Griner, 2012).

The density requirements of UAVs that use CNPC thru the year 2030 were also specified (NASA, 2012). Table 3 shows in the third column the numbers of different sizes of UAVs that can be supported if a terrestrial-based communication link with a 100 km radius is used.

Based on the defined UAV density, the required bandwidth for CNPC links is 90 MHz divided into 34 MHz for the terrestrial-based LOS CNPC links and 56 MHz for the BLOS CNPCs links (Kerczewski and Griner, 2012). Two frequency spectrum ranges were assigned by the 2012 International Telecommunications Union World Radio Communications Conference (WRC-12) for use by CNPC for reliable and real-time data transmission. These frequency spectrums are 960–1164 MHz (L-Band) and 5030–5091 MHz. However, a portion of the first range will be shared with other legacy applications for surveillance and navigation purposes. Another issue of CNPC links is the need for high security requirements. Good security mechanisms should be used for CNPC links to avoid any possibility of spoofed control of navigation signals that may allow unauthorized persons to control the UAVs (Zeng et al., 2016).

For satisfying future volume requirements in aeronautical communication, a new air-ground communication system, called L-Band Digital Aeronautical Communication System (L-DACS), is being developed in Europe with funding from EUROCONTROL. L-DACS is the Future Communication System (FCS) for L-band, 960–1164 MHz. L-DACS includes L-DACS1 (Sajatovic et al., 2009) and L-DACS2 (Fistas, 2009). L-DACS1 is a multi-carrier broadband Orthogonal Frequency-Division Multiplexing (OFDM) based system, while L-DACS2 is a narrow band single-carrier with Gaussian Minimum Shift Keying (GMSK) modulation system. The specifications and advantages of L-DACS1 and L-DACS2 are available in (Neji et al., 2013).

6.3. Licensing and certifications

One of the important subjects that challenge the deployment of UAV applications for smart cities is licensing. UAVs need special certifications to fly in a city's airspace. For example, the US FAA has several

Table 3
CNPC supported aircraft density.

Aircraft size	Density of UAVs in space (km^3)	No. of UAVs within radius of 100 km
Small	0.000802212	1680
Medium	0.000194327	407
Large	0.00004375	91

criteria for UAVs allowed to fly in US airspace. Additionally, for UAVs used in civil applications, getting Airworthiness Certificates in the experimental category is currently the only way civil operators of UAV are accessing NAS. Experimental certificate regulations prevent carrying people or property for compensation or hire. However, they do permit operations for research, development, and training. The FAA is working on preparing and improving the UAV airworthiness certifying process. The process of integrating UAVs into NAS is still undergoing due to the huge number of agencies involved in that process locally and globally such as DoD (US Department of Defense), ICAO (International Civil Aviation Organization), FAA, and NASA. In addition, the FAA decided to provide COA (Certificates of Waiver or Authorization) (FAA, 2014a), which is a certificate that could be available for public firms that want to use UAVs in civil airspace. This certificate is targeting the firms that have UAV applications for law enforcement, firefighting, border patrol, disaster relief, search and rescue, and other government related purposes. Another category is the model aircraft, flying model UAVs is regulated by the FAA Advisory Circular 91-57, which generally restricts operations for hobbies and recreation to below 400 ft to avoid the high-density air traffic areas. In June 2014, the FAA issued a Federal Register notice on its explanation of the legal standards for UAV specifications. The aim of the law is to give the FAA the right to place some actions against model aircraft operators who operate their UAVs in a risky manner. Furthermore, deployment of such standards and regulations is intended to protect users of the airspace and the people and property on the ground (FAA, 2014a) and (FAA, 1981). This could be useful in organizing the simulation and testing of UAV applications and future operations of UAVs.

Furthermore, UAVs have applications related to recreation and hobbies in smart cities and to overcome issues with these, the FAA put some limitations and regulations for such usage. Based on the FAA Modernization and Reform Act of 2012 (FAA, 2014b), a UAV that is utilized for a specific purpose and application is only allowed to fly according to its allocated purpose, mission or task. In addition, the UAV must be operated according to the safety guidelines. The UAV is limited to specific payload certified through a design, construction, inspection, flight test, and operational safety program. In addition, the used UAV should be operated in a way that does not interfere with the local airspace use and the local air traffic. To avoid future risks related to air traffic use when a UAV is within 5 miles of an airport, the operator of the UAV should receive a permission. In the same context, the Australians went beyond that point by implementing license requirements for UAV pilots. A UAV Controller Certificate is required for a pilot to control a UAV according to the UAV Operations Manual. The UAV Controller Certificate is in essence a "Pilot License" and is required by all commercial operators of UAVs or RPA (Remotely Piloted Aircrafts) in Australia (RPAS, 2014).

7. Potential impacts and risks

Employing UAVs for smart cities can positively impact a city's services, infrastructures and residents in different ways. The quality and performance of some smart cities services can be augmented and improved using UAVs. For example, employing UAVs for applications like traffic monitoring and management, merchandise delivery, health emergency services, UAV-aided wireless communications and air taxi services can enhance these services in terms of quality, productivity, timeliness, reliability, and performance. In addition, employing UAVs for applications such as remote health emergency support, agriculture management and environmental monitoring can enhance operations and could help reduce the costs of offering these services.

Employing UAVs can also enhance safety and security and help save human lives in smart cities. UAV applications such as security and crowd monitoring, health emergency services, and large-scale disaster management can significantly contribute in making smart cities safer for residents and visitors. Furthermore, some UAV applications can

Table 4
A summary of UAV applications to smart cities and their potential positive impacts.

UAV application	UAV functions	Potential positive impacts
Traffic monitoring and management	- Provide real-time information about traffic status - Support on ground traffic control components - Search for unoccupied car park spots.	Faster solutions can be applied for different traffic congestion problems, thus helping reduce or solve them.
Health emergency services	- Deliver medical supplies to emergency patients - Act as ambulances and rescue vehicles for remote locations	Saving human lives and reducing the cost of offering health emergency services for remote areas.
Security and crowd monitorings	- Collect real-time information about individuals' activities to support safety and security operations - Monitor crowds in large events	Cities become a safe place for living, events, and businesses.
UAV-based infrastructure inspection	- Collect infrastructure data (e.g. images and scans) - Analyze collected data for identifying and locating problems.	Timely maintenance can be achieved to help avoid major problems that may cause human deaths or/and be very costly to recover from.
Agriculture management and monitoring	- Dispense fertilizers, pesticides, seeds, and water - Monitor the agriculture environment - Help provide mechanisms to protect crops	Enhanced agriculture management process, increased and enhanced productivity, and reduced cost.
Tourism support	- Act as tour guides - Provide mobile security cameras - Offer cameras that visitors can access and use - Provide autonomous aircrafts for tours	Encouraging/increasing tourism. It also improves the image of a city as a good tourism and tech place, which could bring in more external investments.
UAV-based surveying	- Perform 2D Surveying - Perform 3D Surveying - Perform geospatial land surveying	Better management for different types of large construction projects and city management can be achieved.
Large-scale disaster management	- Monitor and communicate disaster status - Search for survivors - Fight fires - Enable communications for emergency activities - Transfer medical supplies and/or the injured	Faster and more effective control and management of large-scale disasters leading to fewer casualties, better control of losses, and faster more targeted response activities.
Merchandise delivery	- Fast delivery of packages	Fast delivery enables better service for customers ordering food and perishable items. New businesses can be initiated due this capability.
UAV taxi	- Transport passengers	People can move within a city faster avoiding ground traffic congestion.
UAV-aided wireless communications	- Provide scalable wireless communications - Function as a backup for faulty wireless nodes - Enable wireless connectivity over distant areas	Wireless communication will be provided in all circumstances. This will enhance the performance of services requiring wireless communication.
UAV-based fog computing	- Support low latency services - Support location aware services - Support geographically widely distributed services - Provide streaming communication and processing - Support mobility and access control - Provide Quality of Services (QoS) provisions - Provide efficient communication with other systems	Enhanced performance of different smart cities services and more effective integration of different services, which will indirectly enhance the quality of life.
Data collection in WSN Using UAVs	- Collect sensors data - Preprocess collected data	Enhanced energy efficiency of some WSN-based smart cities applications. Better delivery of sensed data.

motivate current and new businesses and provide a good image for a city thus helping attract new businesses. Examples include merchandise delivery, tourism support, and air taxi services. These applications can bring more visitors and generate new business ideas thus positively impacting the economy.

Some of the discussed UAV applications have direct impacts on a city residents while others have indirect impacts. For example, merchandise delivery, health emergency services, and air taxi services will provide enhancement for smart cities services that residents directly deal with. In contrast, applications such as UAV-aided wireless communication, UAV-based fog computing, and data collection in WSN can be utilized to enhance different services in a smart city while being hidden from the residents. These applications improve various other areas in a smart city, yet do not have direct interaction with the residents. Table 4 provides a summary of UAV applications, their functions, and potential impacts.

Although there are many positive impacts for applying UAVs for smart cities, there are also some potential risks. Table 5 summarizes the different risks involved with different UAV applications for smart cities. The most important issues for most applications are safety and security. However, the levels of safety and security needed differ from one application to another. For example UAV applications that involve carrying people require extremely strong safety and security measures as it involves high risks of people losing their lives or being injured. Examples of these applications are air taxis and Ambulance UAVs.

In business, as with introducing any new technology, using UAVs in some businesses can put pressure on other similar businesses that use less technological business models. This in some ways provides an unfair advantage for the adopters of the newer technologies like the UAVs. As UAVs start to be more reliable, safe, secure, efficient, and cost effective and more applications are created, businesses may have to adjust their operational models quickly to remain competitive. At the same time, while big businesses can easily adopt UAV solutions for some of their activates, it is difficult (costly) for small business to adopt them. This will negatively impact small businesses as well as some types of jobs.

Despite their costs, most modern and developed countries have a better position to adopt and use smart cities UAV applications. However, this poses great Burdon on resource scarce countries because of the high costs and advanced technologies needed. There are three important aspects for smart cities UAV applications: the technical and non-technical requirements; the benefits of the applications within the context of the smart city; alternative solutions that do not use UAVs and provide comparable replacements. The technical and non-technical requirements of a UAV application define the cost of such application. High security and safety requirements, for example, can add high costs to an application. At the same time, if the UAV application's benefits are limited and there are other cost-effective alternative solutions, then it is not feasible to deploy the UAV application. On the other hand, UAV applications with low technical and non-technical requirements and

Table 5
A summary of issues and potential risks of UAV applications to smart cities.

UAV application	Most challenging issues	Potential risks
Traffic monitoring and management	- High bandwidth and low-latency communication requirements to transfer video streams to the traffic control center.	Generally low risk; however, if security is not strong enough we run into possible major hacking and sabotage incidents.
Health emergency services	- High safety, security, and reliability requirements - High production, development and maintenance costs (more prominent for UAV ambulances)	The possibility of wrong use for the delivered medical elements. There is also the risk on humans in case of malfunctions or crashes
Security and crowd monitorings	- High safety requirements - Fast analysis and observations - Reliable communication with the control center.	UAV accidents may cause some human injuries or deaths. Improper use may lead to ethical issues and legal concerns.
UAV-based infrastructure inspection	- Fast and accurate analysis and observations - High storage needs for inspection applications relying on imaging and video streaming. - Reliable, low latency, and high-bandwidth communication with control centers. - Integration with other systems for more analysis.	Some accidents may cause some damages to the inspected infrastructure. In some cases faulty sensing or delivery of sensed data may lead to incorrect analysis.
Agriculture management and monitoring	- Capability to handle different agriculture situations (pay load, special tools, location, etc.)	Generally low risk, but some errors may cause damages to farms or crops.
Tourism support	- The design of UAVs (safety, accurate operations, etc.) - Tours interaction mechanisms.	Possible risks of accidents causing damages to area or to the tourists. Some privacy concerns may arise when tourists' data is collected.
UAV-based surveying	- Accurate measurement capabilities.	Faults with the measurement devices or methods may lead to wrong surveying results.
Large-scale disaster management	- Reliable communication with the control centers. - Fast and accurate analysis and observations - Capability to deal with different disaster situations. - Advanced coordination if multiple UAVs are used.	Large-scale disasters can take different levels and types of damages. The deployed UAVs may not be able to deal with all situations if there are not designed to handle them. This may lead to inaccurate observations or reduced efficiency.
Merchandise delivery	- Highly efficient operations - Integration with other logistics systems - Capability to handle high weight and large payloads.	Possibility of accidents that may cause property/merchandise damages or injuries.
UAV taxi	- Capability to carry high weights - High safety, security, and reliability. - High levels of autonomization - High development and maintenance costs - Legal aspects related to any accidents.	Any accident may lead to the possibility of injuries or death. Insurance issues and legal consequences may arise. In addition, any accident may reflect negatively on the reputation of the city itself.
UAV-aided wireless communications	- Optimization of UAV positioning and flying patterns for better coverage - Need for energy efficiency	Generally low risk, some issues with safe operations and security may arise.
UAV-based fog computing	- Location selection and load balancing among multiple fog nodes - Integration with other systems - Energy efficiency	Generally low risk if operated within defined parameters.
Data collection in WSN Using UAVs	- High security requirements for critical applications. - Efficient path planning - Integration with other systems - Need for energy efficiency	Risks related to high delays in collecting critical data leading to further delays in the decision making.

high benefits become feasible and approachable even in countries with limited resources and constrained economies.

For example, the UAV Taxi application involves high security, reliability, and safety requirements which can add high costs to the application. Therefore, a UAV taxi is not feasible in all cities. It is only feasible in large cities that can highly benefit from this alternative transportation method. In such case an alternative may exist in the form of piloted helicopter taxis, yet in comparison, it is a lot more expensive. However, other UAV applications such as UAV-based large-scale disaster management and UAV-based environmental monitoring can have low technical and non-technical requirements. These applications can provide high benefits while there are no good cost-effective alternative solutions for them. This makes utilizing these applications in all countries including resource scarce countries very feasible.

8. Conclusion

In the near future many technologies will change the world and transform the economy. Technologies include the Internet of Things, robotics and autonomous systems like UAVs, and cloud computing among many others have evolved significantly. Thus leading to great opportunities to smartly utilize them for a vast set of advanced, autonomous and intelligent applications in almost every field and domain. One vision that is taking place is creating the smart cities. Integrating

UAVs into smart cities will provide huge opportunities and cost-effective intelligent solutions for many problems. Several of these opportunities will create extremely useful applications in intelligent traffic management, smart safety and security, smart media, smart health care, and smart guidance among many. The majority of these applications and future ones can greatly benefit from (some can only be achieved by) integrating UAVs in smart cities.

As discussed in this paper, Along with the wide range of possible applications using UAVs in smart cities, several open issues need to be considered. Examples of these issues include licensing and certification, communication and security, and UAV and smart cities seamless integration. Various Policies and approaches are under investigation. Governing bodies are already working on introducing regulations, policies and operations guidelines to follow for safe use of UAVs. Researchers are introducing models and architectures to build, integrate and deploy UAV applications. In addition, the industry is scrambling to take advantage of the current developments and introduce value-added features to smart cities using UAVs. With improved development and deployment models, enhanced technologies and more focused efforts in creating such applications, it will soon be possible to see actual and useful applications for UAVs in smart cities. More work is still needed in all areas and some of the possible future work we envision include the design and creation of suitable and efficient development models, enhanced middleware platforms, and better

deployment and integration techniques. These efforts will help provide access to more UAV relevant services and technologies, smooth the development efforts, facilitate seamless integration with other systems, enhance operability, and ultimately fully incorporate UAV applications in smart cities.

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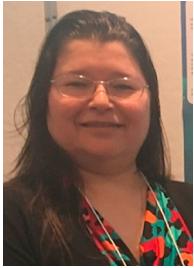
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